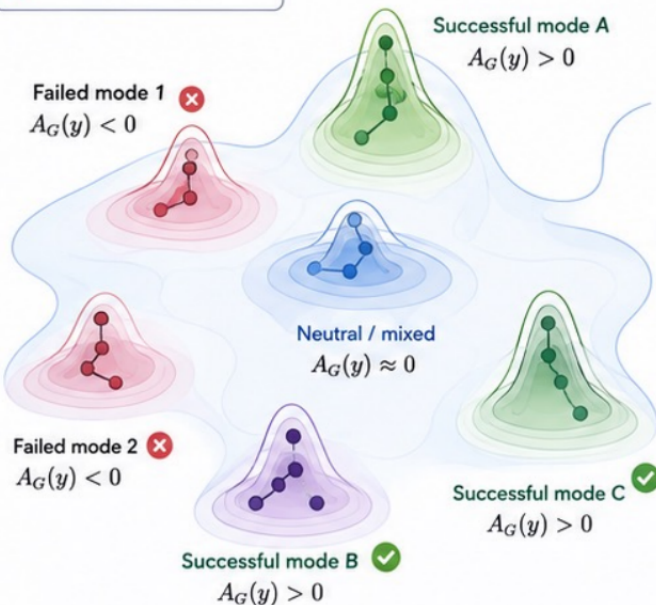


1 Reference policy distribution

$\pi_{\text{ref}}(y | x)$ over trajectories y

✓ Successful (high reward)

✗ Failed (low reward)



Reference policy spans multiple diverse trajectory modes, including both successful and failed behaviors.

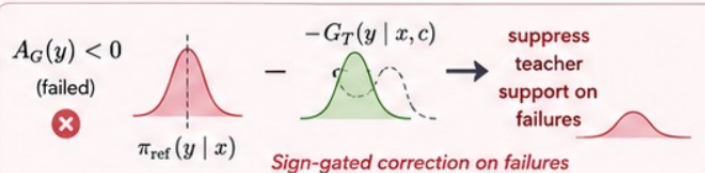
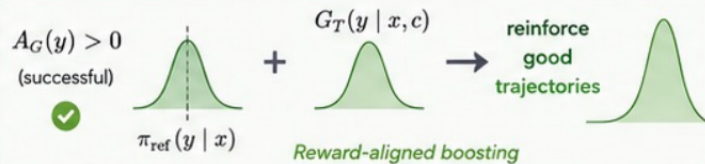
2 FlowBalance energy and sign gating

FlowBalance energy (sign-gated)

$$E(y) = \eta_A A_G(y) + \beta_T G_T(y | x, c) \text{sgn}(A_G(y))$$

How sign gating works

● $A_G(y) > 0$ (good) ● $A_G(y) < 0$ (bad)



Target reweighting

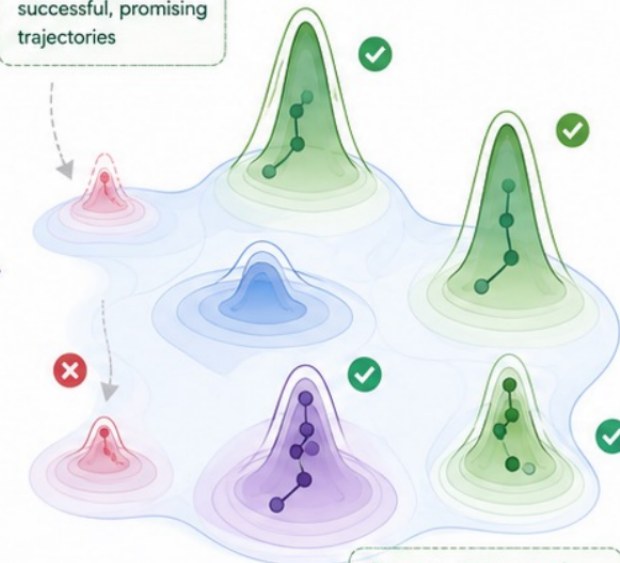
$$p^*(y | x, c) \propto \pi_{\text{ref}}(y | x) \exp\left(\frac{E(y)}{\tau}\right)$$

Temperature τ controls the strength of reweighting.

3 Reweighted target distribution

$p^*(y | x, c)$ over trajectories

Mass increases on successful, promising trajectories



Diversity is preserved: multiple successful modes retain high mass

Higher mass on successful trajectories while maintaining diversity across multiple good modes.



Flow-based target matching

Reweight π_{ref} to obtain a principled target distribution over trajectories.



Sign-gated privileged feedback

Use verifier advantage and teacher signal, flipping teacher support on failures.



Diverse successful trajectory distribution

Boost high-advantage trajectories while retaining multiple good solutions.